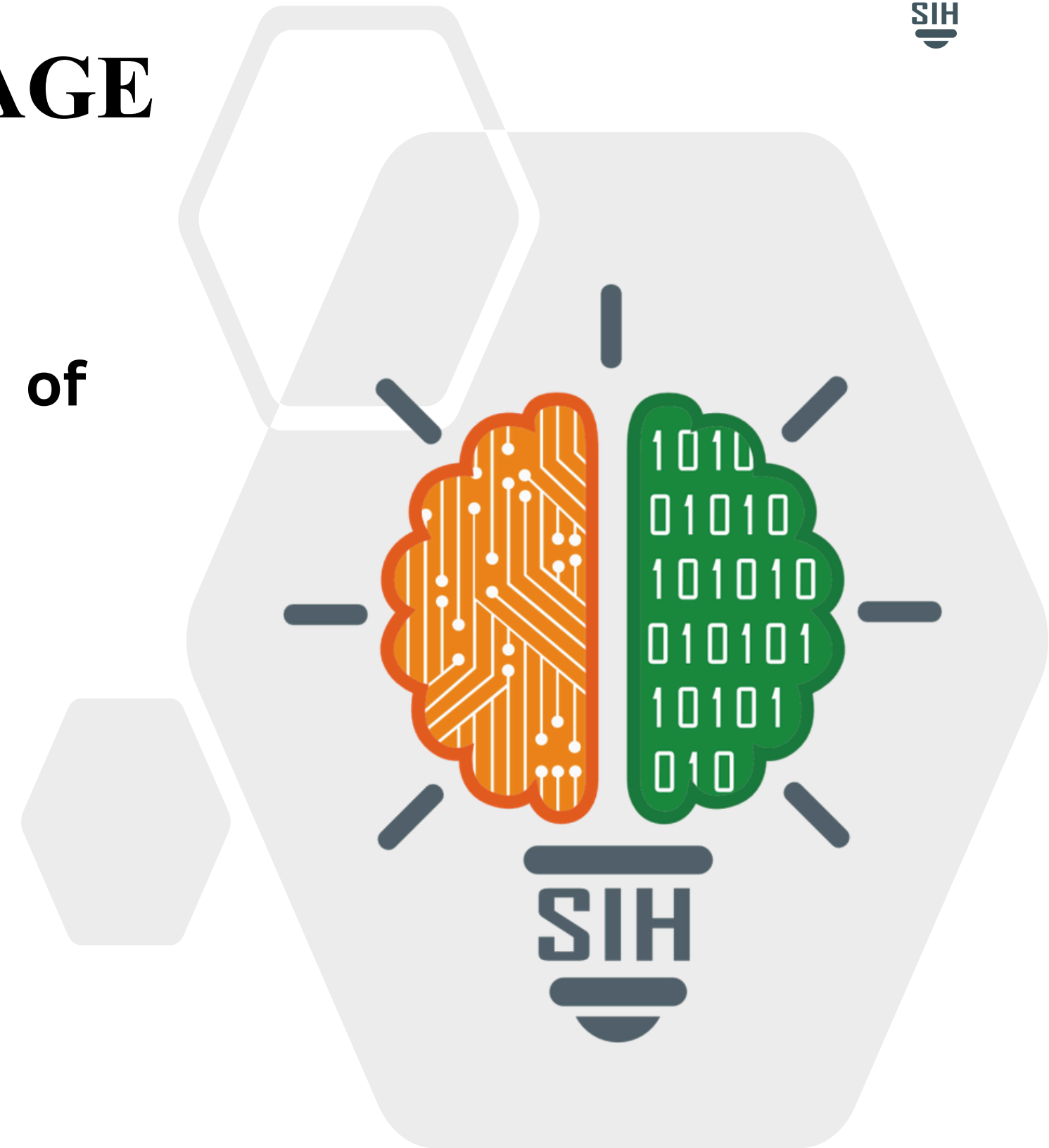


TITLE PAGE

- Problem Statement ID – SIH26133
- Problem Statement Title- Accessibility of public healthcare services in rural areas
- Theme- MedTech / bioTech / HealthTech
- PS Category- Software
- Team ID-
- Team Name - The Bridgers

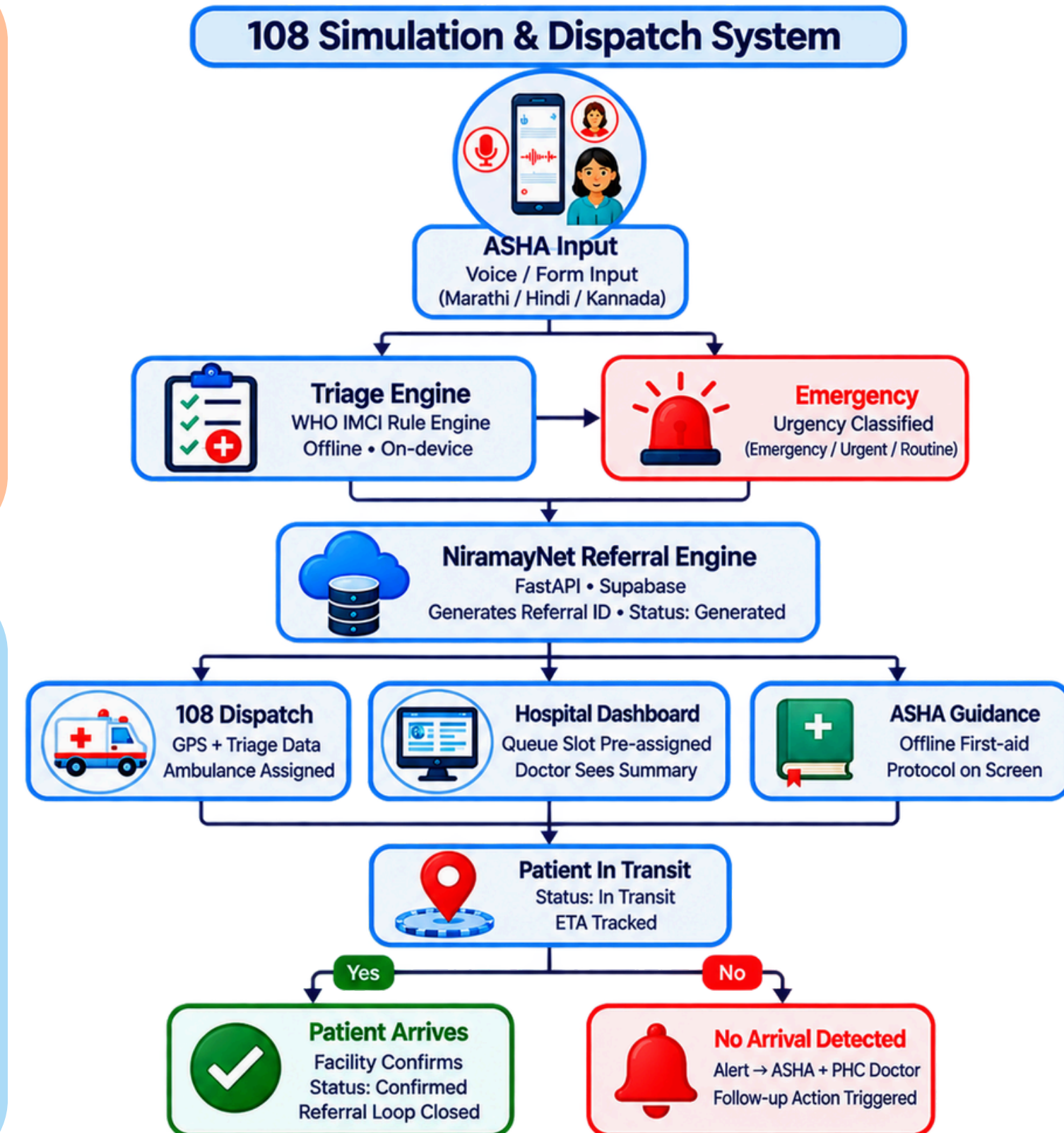


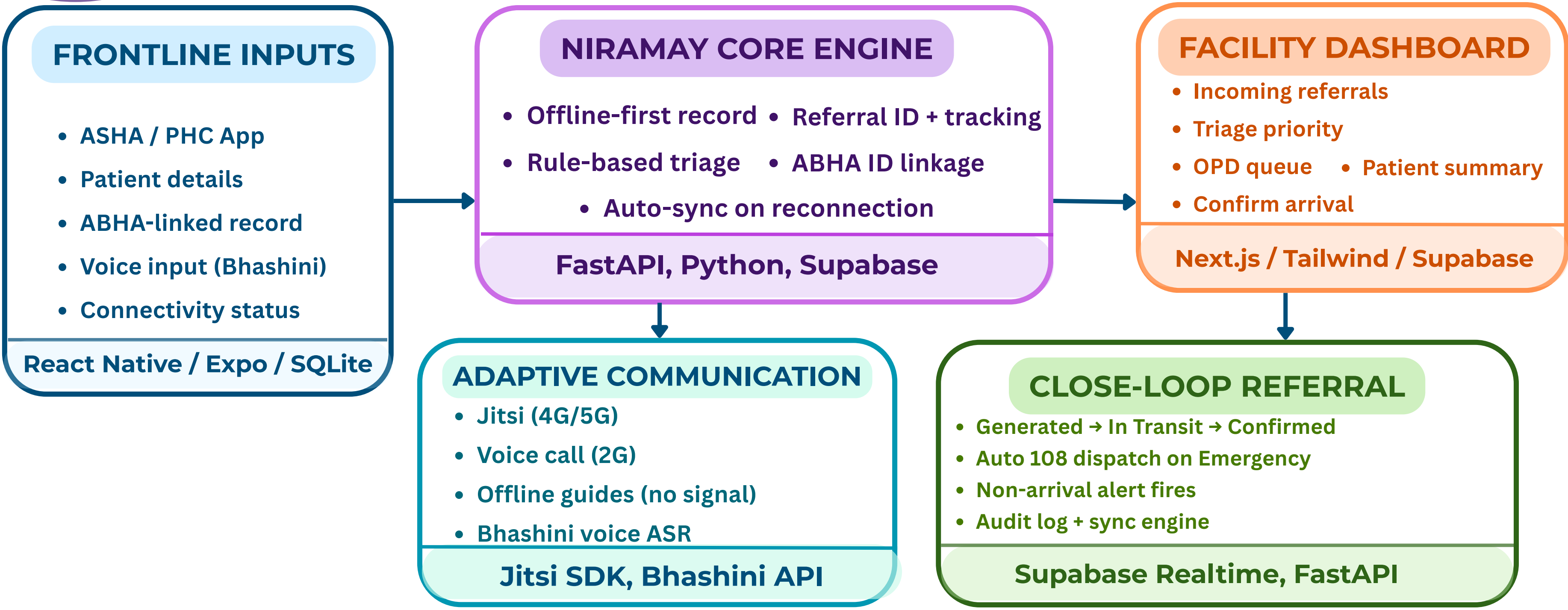
NiramayNet : From Village to Hospital – Unbroken.

- **Intelligent Triage First** – ASHA inputs symptoms via voice (Marathi / Hindi / Kannada). WHO IMCI rule engine classifies Emergency, Urgent or Routine – fully offline, on-device.
- **Adaptive Communication** – Strong signal → Jitsi video. Weak signal → voice call. Zero signal → offline clinical guides. Triggers automatically, no manual switching.
- **Real-Time OPD Queue & Pre-Assigned Slots** – Referral generated → facility dashboard updates instantly → patient has a queue token before leaving the village. Emergency cases flagged at the top automatically.
- **End-to-End Referral Tracking + 108 Integration** – Emergency tier auto-dispatches 108 with patient GPS and triage summary. Referral tracked: Generated → In Transit → Confirmed. No arrival within expected window → automatic alert fires.

Innovation & Uniqueness

- **Referral loop closure** – three-state tracking with automatic non-arrival alerts. No deployed government system confirms whether a referred patient actually arrived.
- **Offline-first by design** – triage and records work at zero signal. Sync happens invisibly on reconnection.
- **Triage gates the referral** – no referral without a completed triage first. Targets the 58% inappropriate specialist referral rate on existing platforms.
- **Strengthens existing infrastructure** – built on ABHA ID, existing ASHA workforce and 108 network. Connective layer, not a replacement.





TECH STACK	Mobile App	Backend	Web Dashboard	Integrations
	React Native / Expo SQLite, NetInfo	FastAPI, Python Supabase	Next.js, Tailwind CSS Supabase Realtime	Bhashini API Jitsi Meet SDK

feasibility

- Offline-first mobile app with local data storage and automatic sync
- Integrates with existing ABDM/ABHA and facility registries
- Modular architecture enables phased deployment across PHCs and districts

viability

- Strengthens existing public-health workflows instead of replacing them
- Reduces referral delays, unnecessary travel and waiting time
- Scalable from pilot PHCs to district/state level using cloud-based backend

FEASIBILITY AND VIABILITY

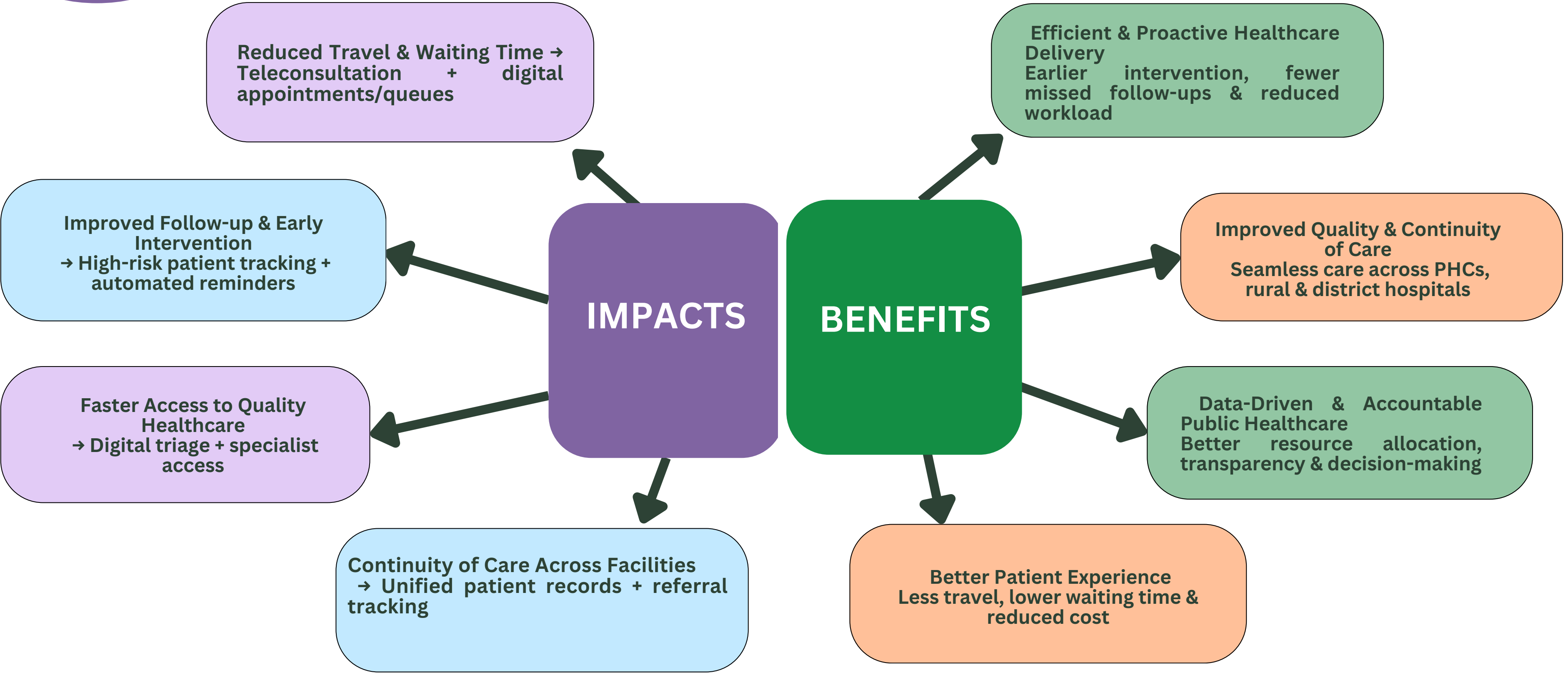
- Low connectivity & device limitations
- Facility data & interoperability gaps
- Adoption, privacy & security

challenges and risks

- Offline auto-sync
- Protocol-driven triage & role-based access
- Phased deployment & system integration

strategies

IMPACTS AND BENEFITS



- **NHM – 16th Common Review Mission (2024)** : Identifies gaps in rural referral mechanisms, follow-up and continuity of care.
Link - [NHM – 16th Common Review Mission 2024](#)
- **I-TREC Program – India (2020)** : Demonstrates electronic referral tracking and coordinated care across India's public healthcare system
Link - [PubMed - I - TREC program](#)
- **Nikshay – Electronic Referral & Feedback Mechanism (2021)**
Highlights challenges in referral tracking and feedback in India's healthcare system.
Link - [PubMed - Nikshay referral & Feedback Study](#)
- **Rural Referral Pathways – PLOS ONE (2022)** : Examines referral coordination, communication and documentation problems in a low-resource setting.
Link - [PLOS ONE - RURAL Referral Pathways Study](#)
- **WHO – Referral & Counter-Referral Framework** : Provides standardized referral and counter-referral forms for communication and continuity of care.
Link - [WHO - Standardized Clinical Forms](#)